Fuel Prices

Pawan

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Load the necessary packages

```
library(data.table)            ## Raed data table
library(ggplot2)               ## Graphical represantation
library(readxl)
library(gridExtra)
library(stringr)               ## Work with string
library(lubridate)             ## date
library(scales)                ## Nice Scales for ggplot2
library(fontawesome)           ## Icons display in ggplot2
library(ggtext)                ## Markdown text support for ggplot2
library(showtext)              ## Display fonts in ggplot2
library(colorspace)            ## Lighten and Darken colours
library(seecolor)              ## To print and view colours
library(patchwork)             ## Combining plots
```

Costum the plot theme

```
my_theme <- theme_bw(base_size = 16) +  ## Start with theme_bw and increase base size
    theme(text = element_text(family = "serif", color = "#333333"), ## Set font family and c
          plot.title = element_text(size = 20, face = "bold",
                                    hjust = 0.5, margin = margin(10, 0,
                                                                 10, 0)), ## Center and styl
          axis.title = element_text(size = 18),
          axis.text = element_text(size = 14),
          legend.position = "bottom",
          legend.title = element_blank(),  ## Remove legend title
          panel.grid.major = element_line(color = "#EEEEEE"), ## Lighter grid lines
          panel.border = element_blank(), ## Remove panel border
          axis.line = element_line(color = "#333333"), ## Add axis lines
```

```
          axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1),
          strip.background = element_rect(fill = "#F2F2F2",
                                          color = NA) ## Style facet strips (if used)
          )
```

International Fuel Price

Read the data

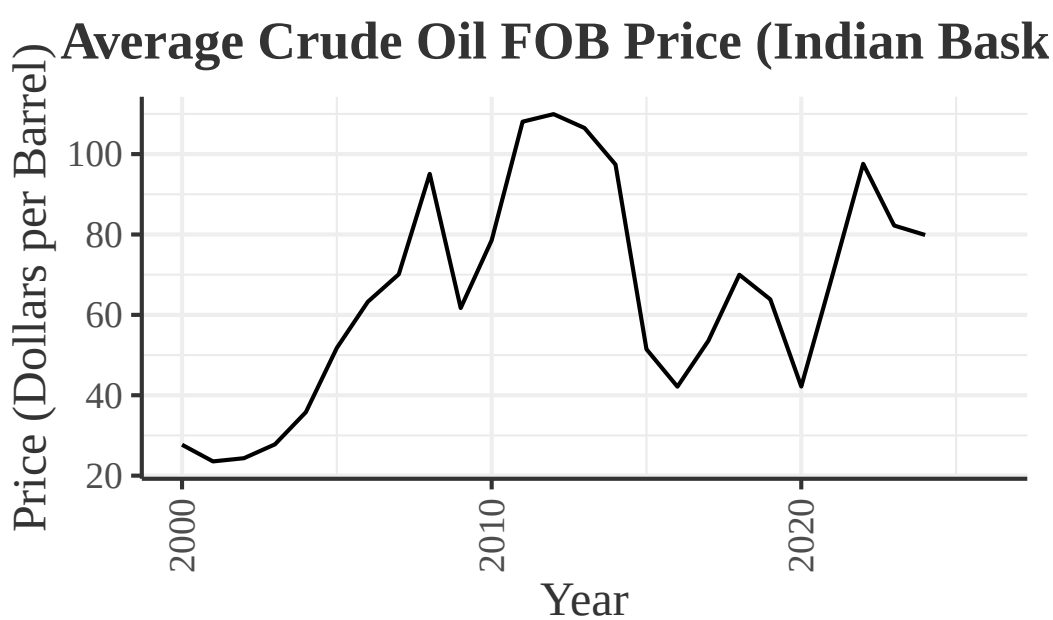
```
as.data.table(read_excel("/home/pawan/fuelprices/international_price_indian_basket.xlsx",
                          sheet = "crude-oil"))->crude

melt(crude, id = c("Year", "Average", "Ratio"))->crude
as.numeric(match(substr(crude$variable, 1, 3), month.abb))->crude$variable
as.numeric(ifelse(crude$variable>3, substr(crude$Year, 1, 4),
                  paste0(20,substr(crude$Year, 6, 7))))->crude$Year

as.Date(paste0(crude$Year, "-", crude$variable, "-", 20))->crude$Date
crude[,.(Price = mean(value)),.(Year)]->avgcrude

## Diesel Internationl Price

as.data.table(read_excel("/home/pawan/fuelprices/international_price_indian_basket.xlsx",
                          sheet = "diesel"))->diesel

melt(diesel, id = c("Year", "Average"))->diesel
as.numeric(match(substr(diesel$variable, 1, 3), month.abb))->diesel$variable
as.numeric(ifelse(diesel$variable>3, substr(diesel$Year, 1, 4),
                  paste0(20, substr(diesel$Year, 6, 7))))->diesel$Year

as.Date(paste0(diesel$Year, "-", diesel$variable, "-", 20))->diesel$Date
paste("Diesel")->diesel$Fuel

## Pertol International Price
as.data.table(read_excel("/home/pawan/fuelprices/international_price_indian_basket.xlsx",
                          sheet = "petrol"))->petrol

melt(petrol, id = c("Year", "Average"))->petrol
as.numeric(match(substr(petrol$variable, 1, 3), month.abb))->petrol$variable
as.numeric(ifelse(petrol$variable>3, substr(petrol$Year, 1, 4),
                  paste0(20,substr(petrol$Year, 6, 7))))->petrol$Year
```

```
as.Date(paste0(petrol$Year, "-", petrol$variable, "-", 20))->petrol$Date
paste("Petrol")->petrol$Fuel

## Merge Diesel and Petrol
rbind(diesel, petrol)->fuel
```

Visualizations

Average Crude Oil FOB Price (Indian Basket)

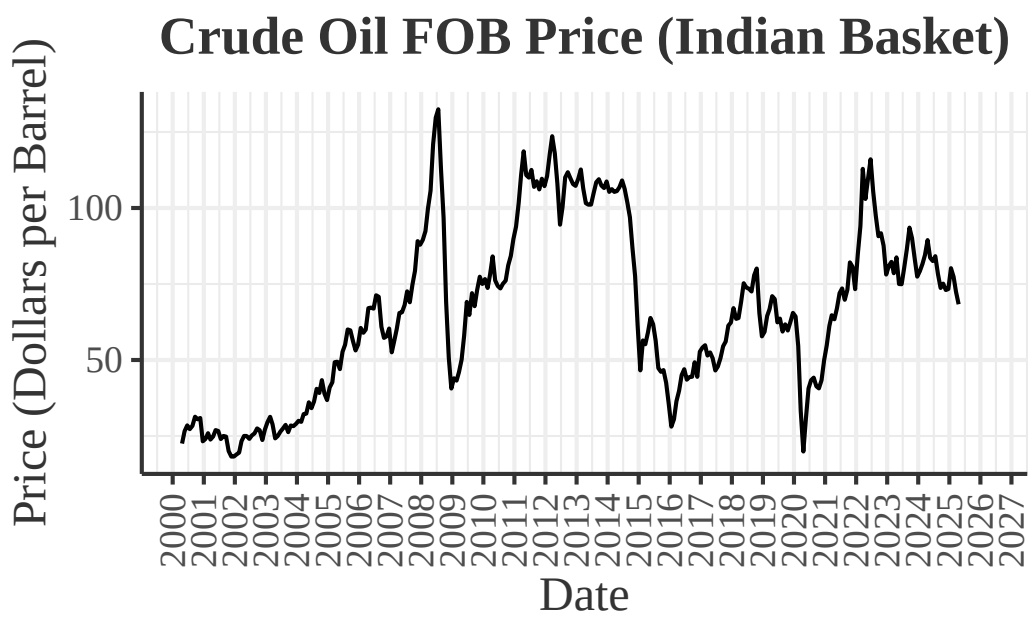
```
ggplot(avgcrude, aes(x = Year, y = Price, group = 1)) +
    geom_line() +
    my_theme +
    labs(y = "Price (Dollars per Barrel)") +
    ggtitle("Average Crude Oil FOB Price (Indian Basket)")
```



Crude Oil FOB Price (Indian Basket)

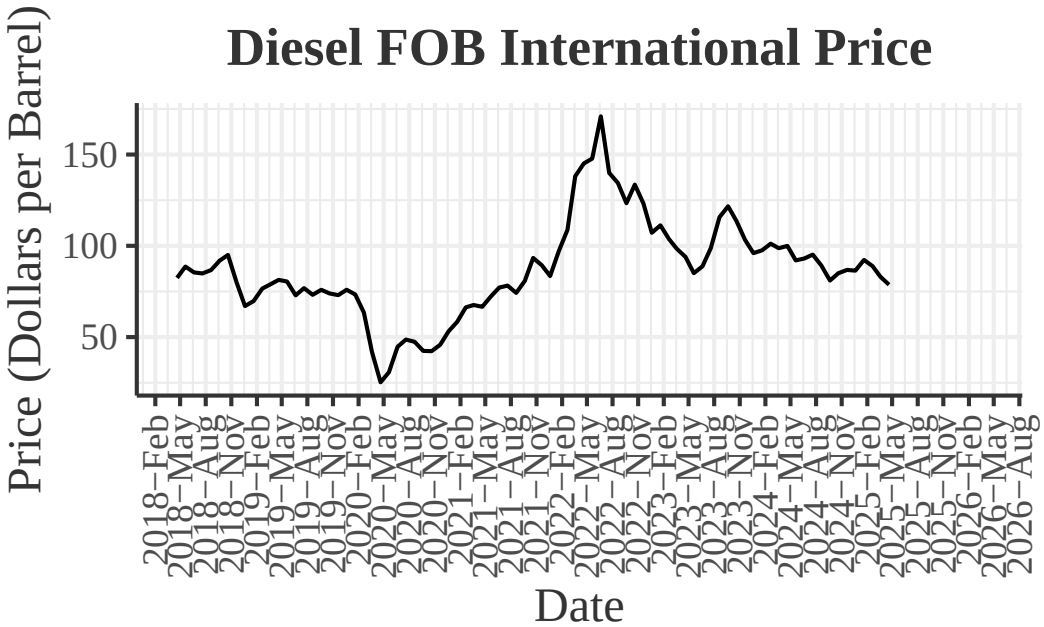
```
ggplot(crude, aes(x = Date, y = value, group = 1)) +
    geom_line() +
    my_theme +
    scale_x_date(date_labels = "%Y", date_breaks = "1 years") +
```

```
    labs(y = "Price (Dollars per Barrel)") +
    ggtitle("Crude Oil FOB Price (Indian Basket)")
```



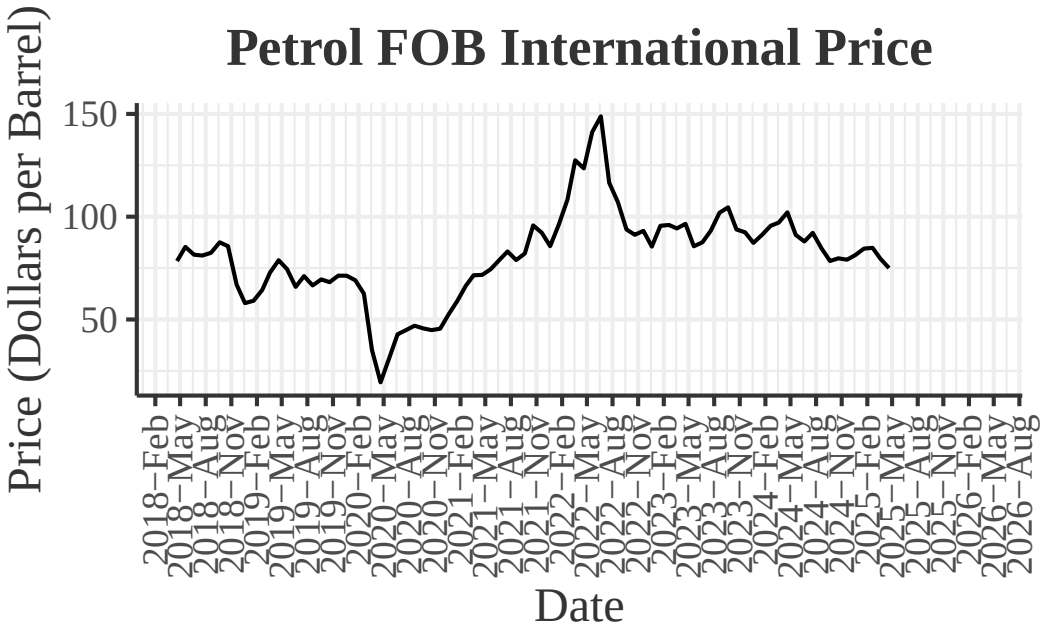
Diesel FOB International Price

```
ggplot(diesel, aes(x = Date, y = value, group = 1)) +
    geom_line() +
    my_theme +
    scale_x_date(date_labels = "%Y-%b", date_breaks = "3 months") +
    labs(y = "Price (Dollars per Barrel)") +
    ggtitle("Diesel FOB International Price")->p1
p1
```

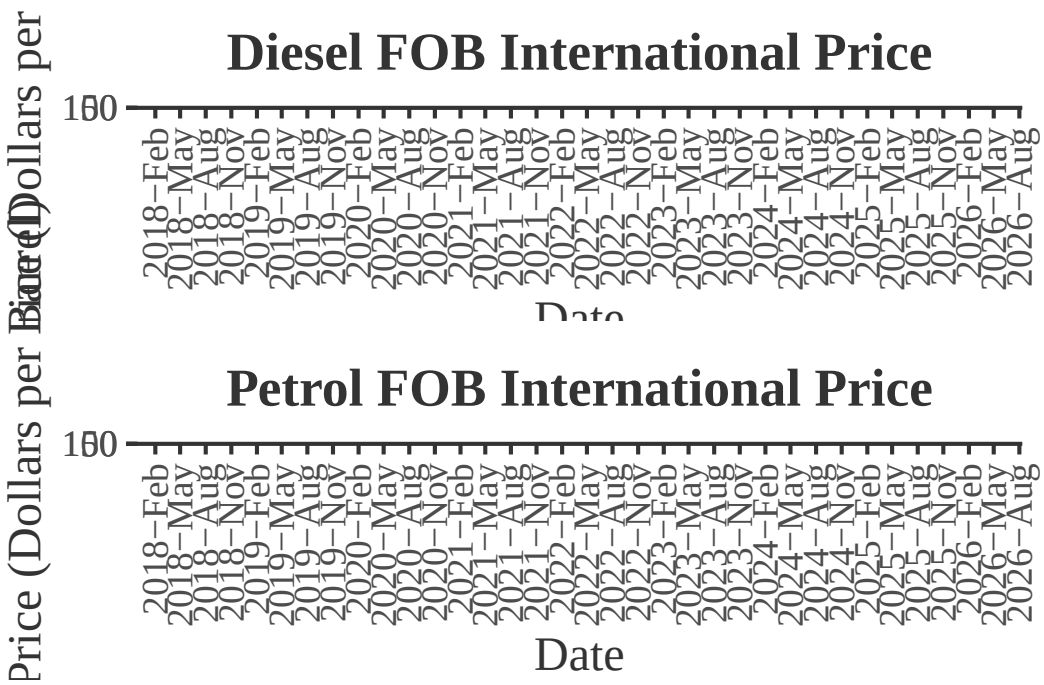


Petrol FOB International Price

```
ggplot(petrol, aes(x = Date, y = value, group = 1)) +
    geom_line() +
    my_theme +
    scale_x_date(date_labels = "%Y-%b", date_breaks = "3 months") +
    labs(y = "Price (Dollars per Barrel)") +
    ggtitle("Petrol FOB International Price")->p2
p2
```

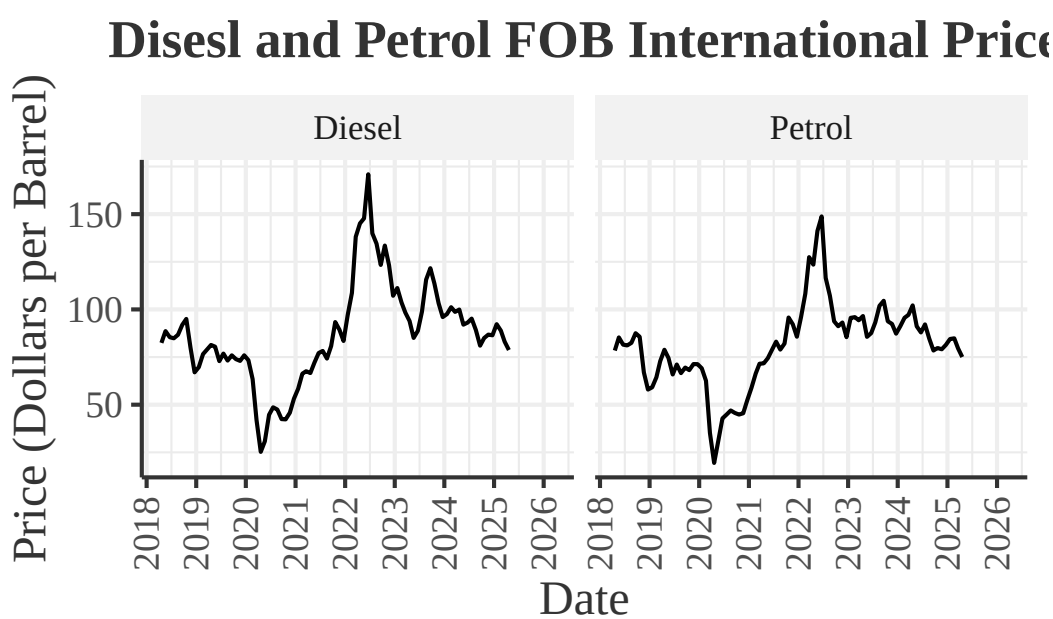



```
grid.arrange(p1, p2)
```



Disesl and Petrol FOB International Price

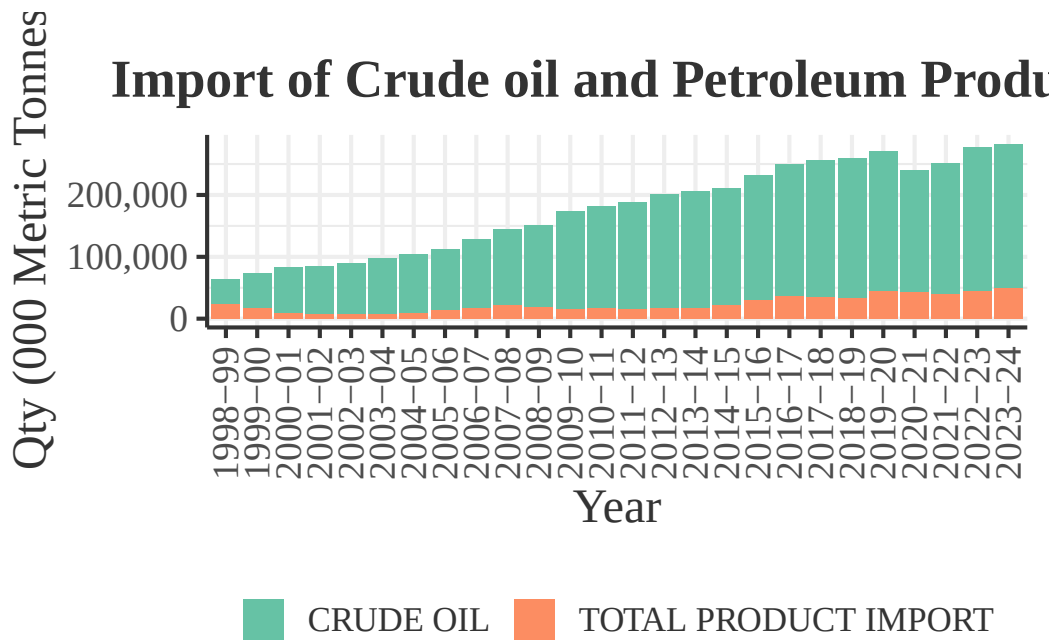
```
ggplot(fuel, aes(x = Date, y = value, group = 1)) +
    geom_line() +
    my_theme + facet_wrap(~Fuel) +
    scale_x_date(date_labels = "%Y", date_breaks = "1 years") +
    labs(y = "Price (Dollars per Barrel)") +
    ggtitle("Disesl and Petrol FOB International Price")->p
p
```



Import of Crude oil and Petroleum Products

```
as.data.table(read_excel("/home/pawan/fuelprices/IMPORT_EXPORT.xlsx",
                         sheet = "PT_IMPORT_H"))->t
melt(t, id = "IMPORT_EXPORT")->t

t[IMPORT_EXPORT %in% c("CRUDE OIL", "TOTAL PRODUCT IMPORT")]->imp

ggplot(imp, aes(x = variable, y = value, fill = IMPORT_EXPORT)) +
    geom_bar(stat = "identity", position = "stack") +
    my_theme +
    scale_fill_brewer(palette = "Set2") +
    scale_y_continuous(labels = scales::comma) +
    labs(x = "Year",
```

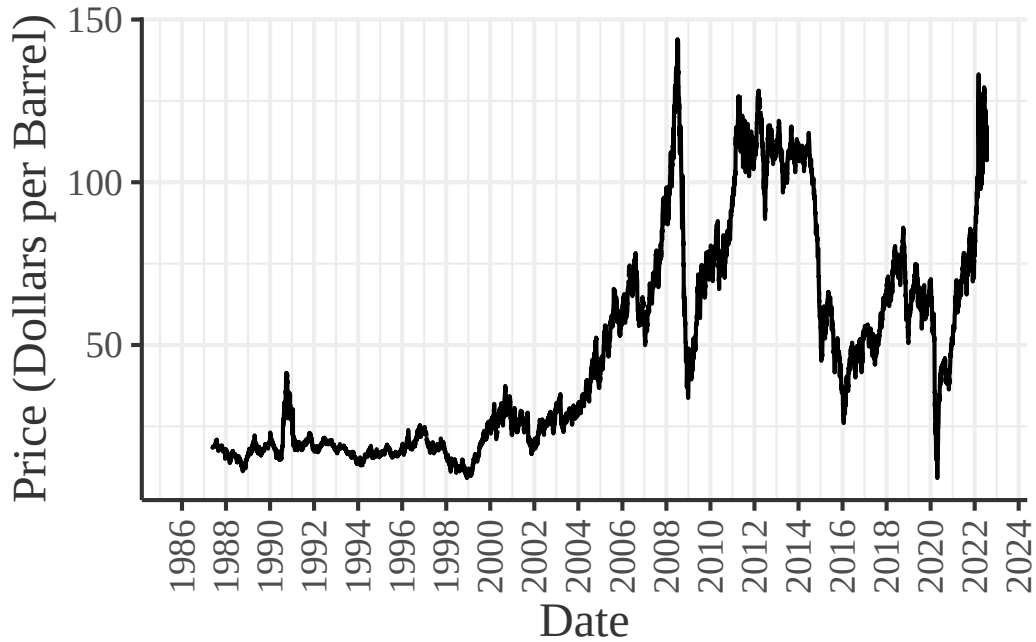
```
        y = "Qty (000 Metric Tonnes)") +
    ggtitle("Import of Crude oil and Petroleum Products")
```




```
##ggsave("import-crude-and-petroleum.png", width = 15, height = 8)
```

Crude Oil Price

```
fread("/home/pawan/fuelprices/eia-oil-prices.csv")->t
as.Date(t$Date,"%Y-%b-%d")->t$Date

ggplot(t, aes(x = Date, y = RBRTE, group = 1)) +
    geom_line() +
    my_theme +
    labs(y = "Price (Dollars per Barrel)") +
    scale_x_date(date_labels = "%Y", date_breaks = "2 years")->p
p
```

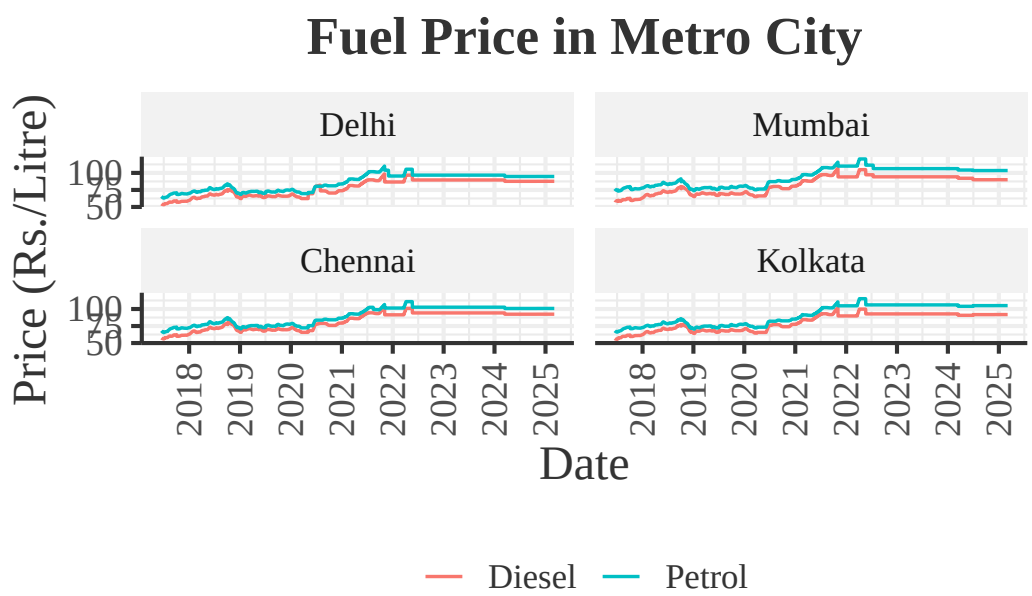


Fuel Price in Metro Cities of India

Load Data

```
fread("/home/pawan/fuelprices/india_retail_fuelprices.csv")->t
as.Date(t$Date, format = "%d/%m/%y")->t$Date

melt(t, id = c("Date", "Fuel"))->t1

t[Fuel == "Petrol"]->petrol
t[Fuel == "Diesel"]->diesel

melt(petrol, id = c("Date", "Fuel"))->petrol
petrol[order(Date)]->petrol
tail(petrol, n = 4)->petrol1

melt(diesel, id = c("Date", "Fuel"))->diesel
diesel[order(Date)]->diesel
tail(diesel, n = 4)->diesel1
```

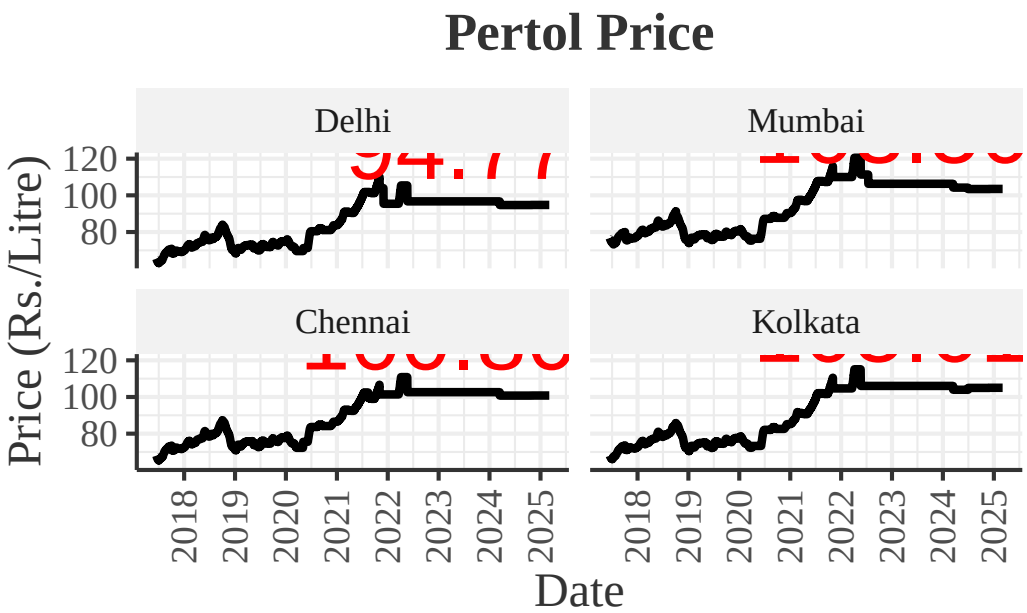
```
ggplot(t1, aes(x = Date, y = value, color = Fuel)) +
    geom_line(linewidth = 0.6) +
    facet_wrap(~variable) +
    my_theme +
    scale_x_date(date_labels = "%Y", date_breaks = "1 years") +
    labs(y = "Price (Rs./Litre)") +
    ggtitle("Fuel Price in Metro City")->p

p
```



Petrol Price

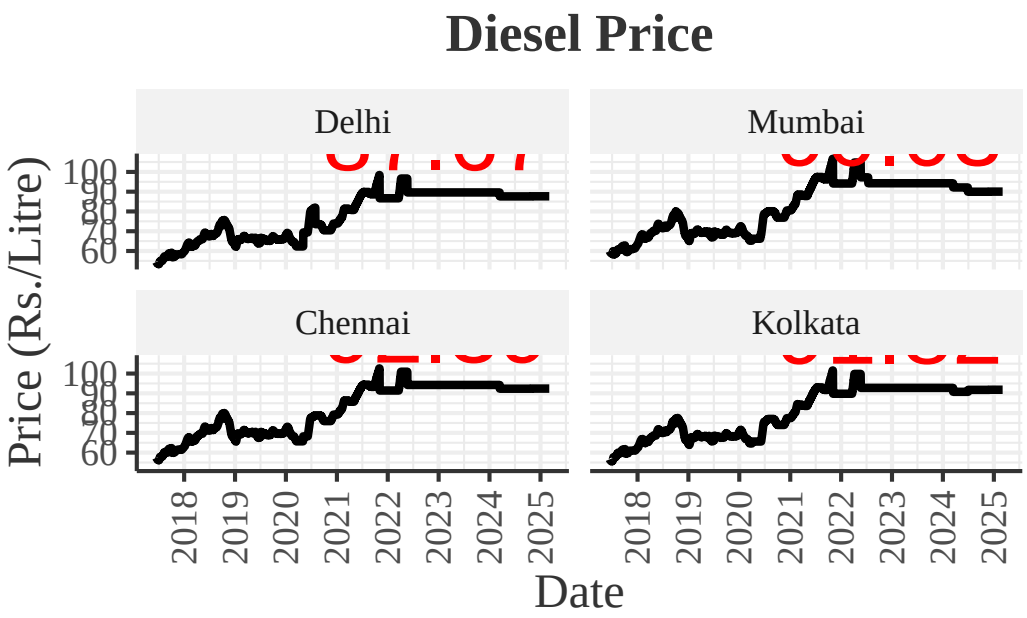
```
## Petrol Price
ggplot(petrol, aes(x = Date, y = value, group = 1)) +
    geom_line(linewidth = 1.5) +
    facet_wrap(~variable) +
    geom_text(data = petrol1, aes(x = Date, y = value,
                                  label = sprintf("%0.2f", value)),
              color = "red",
              vjust = -0.4,
              hjust = 0.9,
              size = 3*4) +
    my_theme +
```

```
    scale_x_date(date_labels = "%Y", date_breaks = "1 years") +
    labs(y = "Price (Rs./Litre)") +
    ggtitle("Pertol Price")->p1
p1
```

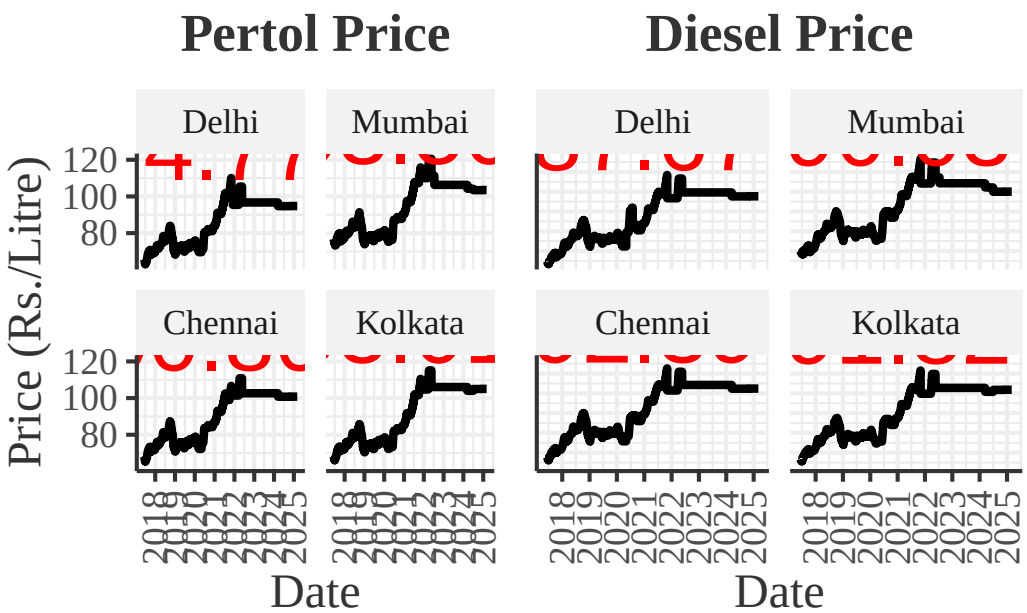


Diesel Price

```
## Diesel Price
ggplot(diesel, aes(x = Date, y = value, group = 1)) +
    geom_line(size = 1.5) +
    facet_wrap(~variable) +
    geom_text(data = diesel1, aes(x = Date, y = value,
                                  label = sprintf("%0.2f", value)),
              color = "red",
              vjust = -0.4,
              hjust = 1,
              size = 3*4) +
    my_theme +
    scale_x_date(date_labels = "%Y", date_breaks = "1 years") +
    labs(y = "Price (Rs./Litre)") +
    ggtitle("Diesel Price")->p2
p2
```



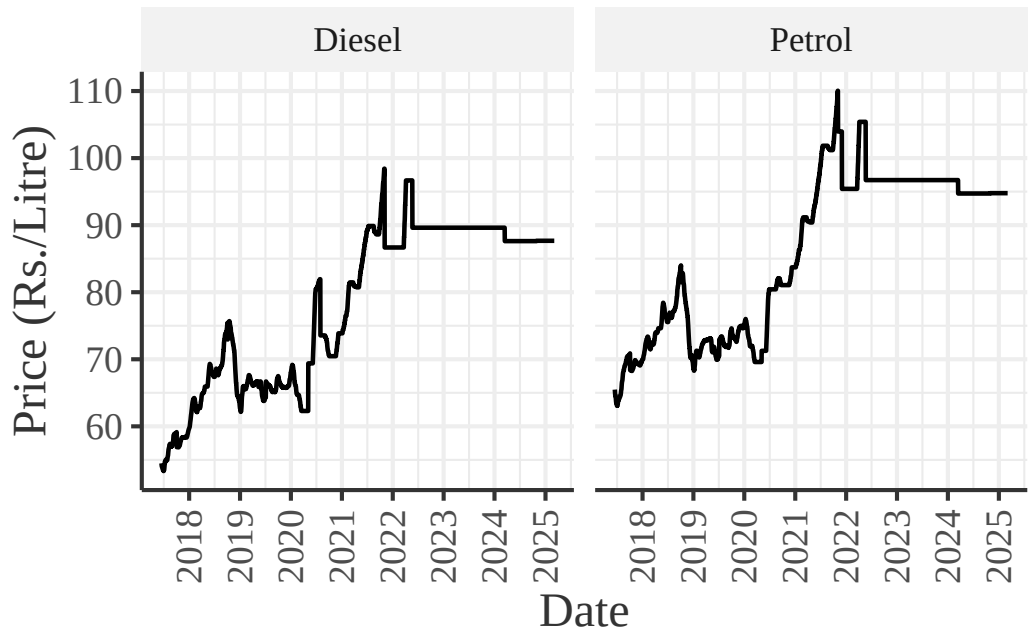

```
cowplot::plot_grid(p1,
                   p2 + theme(axis.text.y = element_blank(),
                              axis.ticks.y = element_blank(),
                              axis.title.y = element_blank()))
```



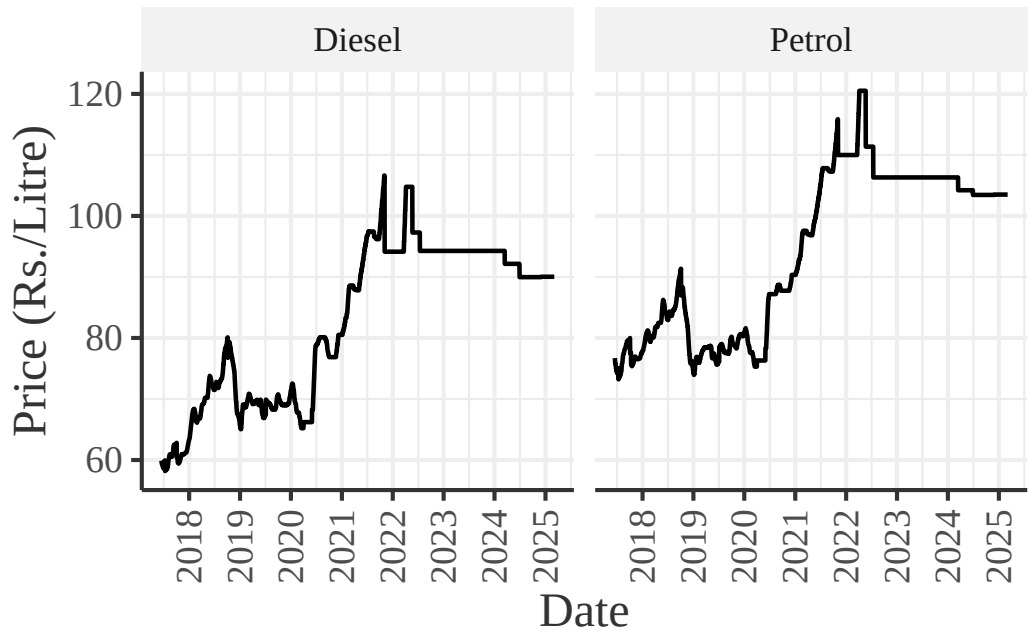

```
## ggsave("fuel-price.png", width = 15, height = 8)
```

Daily Fuel Price in different City

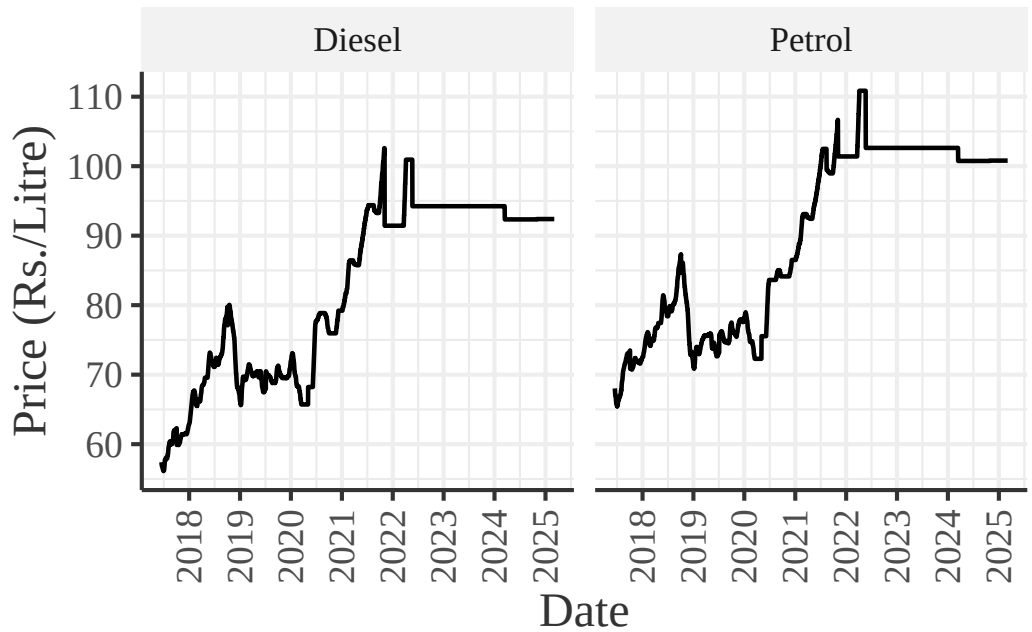
```
## Delhi Petrol Diesel Daily Price
ggplot(t, aes(x = Date, y =Delhi, group = 1)) +
    geom_line(size = 0.8) +
    facet_wrap(~Fuel) +
    my_theme +
    scale_x_date(date_labels = "%Y", date_breaks = "1 years") +
    labs(y = "Price (Rs./Litre)")->p
p
```



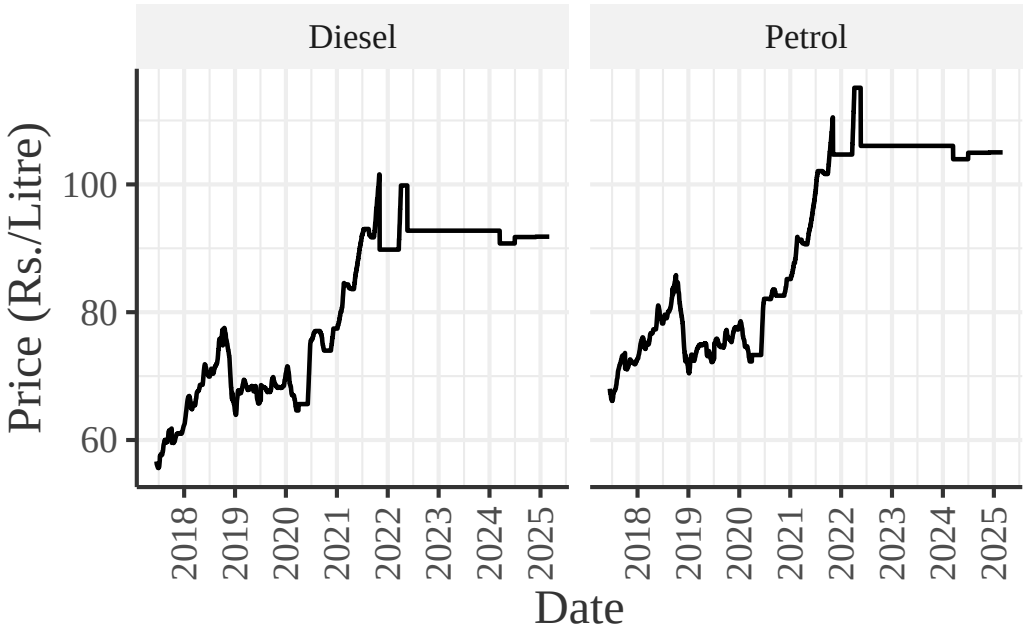

```
## Mumbai Petrol Diesel Daily Price
ggplot(t, aes(x = Date, y = Mumbai, group = 1)) +
    geom_line(size = 0.8) +
    facet_wrap(~Fuel) +
    my_theme +
    scale_x_date(date_labels = "%Y", date_breaks = "1 years") +
    labs(y = "Price (Rs./Litre)")->p

p
```




```
## Chennai Petrol Diesel Daily Price
ggplot(t, aes(x = Date, y = Chennai, group = 1)) +
    geom_line(size = 0.8) +
    facet_wrap(~Fuel) +
    my_theme +
    scale_x_date(date_labels = "%Y", date_breaks = "1 years") +
    labs(y = "Price (Rs./Litre)")->p
p
```




```
## Kolkata Petrol Diesel Daily Price
ggplot(t, aes(x = Date, y = Kolkata, group = 1)) +
    geom_line(size = 0.8) +
    facet_wrap(~Fuel) +
    my_theme +
    scale_x_date(date_labels = "%Y", date_breaks = "1 years") +
    labs(y = "Price (Rs./Litre)")->p
p
```


Diesel
Petrol
Price (Rs./Litre)
100
80
60
2018
2019
2020
2021
2022
2023
2024
2025
Date